

Customer Churn Analysis & Retention Strategy

Executive Summary

Project Overview

This project focuses on analyzing customer churn behavior using data analytics and predictive modeling techniques. The objective was to identify the key factors influencing customer churn and provide data-driven recommendations to improve customer retention and business performance.

The analysis was conducted using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn on a customer churn dataset containing customer demographic, billing, and subscription information.

Business Problem

Customer churn is a major challenge for subscription-based businesses because losing customers directly impacts revenue and growth. The company wanted to understand:

- Why customers are leaving
- Which customer groups are most likely to churn
- How churn can be reduced through strategic business actions

Key Findings

1. Customer Tenure is the Strongest Churn Indicator

Correlation analysis showed that tenure has the strongest negative relationship with churn (-0.51). Customers with shorter tenure are significantly more likely to leave the company.

2. Contract Type Impacts Retention

Customers with long-term contracts demonstrated lower churn rates compared to customers on flexible or short-term contracts.

3. Higher Monthly Charges Increase Churn Probability

Customers with relatively higher monthly charges showed slightly higher churn behavior.

4. Predictive Modeling Successfully Identified Churn Drivers

A Logistic Regression model was developed to predict churn probability and identify important customer retention factors.

Business Recommendations

1. Introduce loyalty programs targeting new customers.
2. Encourage long-term subscriptions through discounts and offers.
3. Improve onboarding experience for first-time customers.
4. Provide retention offers to high monthly charge customers.
5. Use predictive analytics to proactively identify high-risk customers.

Expected Business Impact

Implementing the proposed retention strategies can help the business:

- Reduce customer churn
- Improve customer satisfaction
- Increase long-term customer retention
- Improve recurring revenue stability